

# CIVL4340 Assignment 2

## *Wind tunnel report*

### Overview

This assignment involves analysis of wind tunnel results measured during practicals 1 and 2 and a written report. It is to be done in groups with one submission per group. Within the practicals you will have measured the fluctuating pressures on a model building with and without surrounding models for the purpose of determining pressure coefficients on the body. For the assignment you must report these results and you are also required to use AS/NZS1170.2 to determine the external aerodynamic shape factor,  $C_{shp}$ , on your body. Wind tunnel and AS/NZS1170.2 pressure distributions should be compared and any differences identified and discussed.

### Data analysis

The analysis you are expected to report in your assignment is detailed below.

#### *Fluctuating wind measurements*

Each group is required to:

- Develop mean wind velocity, peak wind velocity and turbulence intensity profiles for the two wind tunnel configuration tested in Practical 1. Ensure you filter your velocity data before calculating your peak value to ensure it is representative of an equivalent 0.2-second peak gust. Think about whether your peak velocity profile is accurate and consider also plotting a statistical peak profile as well.
- Compare peak velocity and turbulence intensity profiles with those provided in AS/NZS1170.2 and determine what terrain category each configuration most closely represents.
- Calculate the along-wind spectra near your building reference height for the tunnel configuration tested in Practical 2. Compare this spectra with the non-dimensional von Karmen spectra discussed in class.

#### *Pressure measurements*

After filtering and segmenting the data you measured in Practical 2 so that you can determine peak 0.2-second (full-scale) pressures within a 10-minute full-scale period, each group is required to:

- Generate a table (and figure, if you like) that shows the mean, maximum and minimum measured pressure coefficients at all model taps for both the isolated building case and the case with surrounding buildings.
- Determine the ratio of pressure coefficients calculated for your two building configurations. Discuss how the surrounding buildings influence pressures on your

model building identifying where pressure coefficients are increased or decreased positing reasons for why changes may exist.

Each group is then required to:

- f) Use AS/NZS1170.2 to determine the external aerodynamic shape factor,  $C_{shp}$ , and pressure coefficients,  $C_{p,e}$  distribution over your building assuming you are providing this information for cladding design where the tributary area of each fixing is less than  $1 \text{ m}^2$  in full scale.
- g) Analyse and plot measured pressure coefficients (based on an appropriate pressure and reference velocity) on your model building and compare these data with AS/NZS1170.2 derived values, with and without your surrounding buildings. Discuss any differences.

## Submission

Each group is required to submit a report summarising your analysis and results. This should be written as though it is a wind tunnel report to be provided to a structural engineer so they can do their façade cladding design. As a guide, your report should include,

- A contribution statement, signed by all group members.
- An executive summary (<1 page) providing a table of your key results.
- A brief introduction (<0.5 page) explaining what tests you have undertaken and why.
- An experimental setup section detailing the wind tunnel, model, pressure taps and data acquisition information (1-2 page).
- A methodology section that outlines what analysis techniques you have used.
- A results section that provides your analysis results and any discussion required. This section should also include your AS/NZS1170.2 distributions.
- Appendices can be included if required. These may include large tables or figures showing pressure tap layout or other data that does not fit neatly into your report.

Your submission is due at **4 pm on Friday May 22**. Please submit your assignment through the link in the Assignment 2 folder on Blackboard. One submission is required per group. Assignment 2 is **30%** of your total mark for the course. The course lecturer will mark the assignment based on the accuracy of your results, quality of figures/tables and overall clarity of your report. A single mark will be awarded for each group submission.